Significant Earthquake Analysis

Phone Kyaw

We will talk about

- Dataset
- Spatial Analysis
- Graph Analysis

Dataset

Significant Earthquakes, 1900-Present

Comprehensive Earthquake Dataset, Magnitude 5+ Since 1900 - Updated Regularly



Data Card | Code (8) | Discussion (1) | Suggestions (0)

About Dataset

This dataset provides comprehensive information on significant earthquakes that have occurred around the world since 1900 with a magnitude of 5 or above. The data includes essential details such as location, date and time, magnitude, depth, and other relevant information about each earthquake.

The dataset is updated weekly and sourced from the United States Geological Survey (USGS), which maintains a global catalog of earthquake information. The dataset includes earthquakes from all regions of the world, from the most seismically active regions like the Pacific Ring of Fire to less active regions like Europe and Africa.

Usability
9.41

License
CC0: Public Domain

Expected update frequency
Weekly

Tags

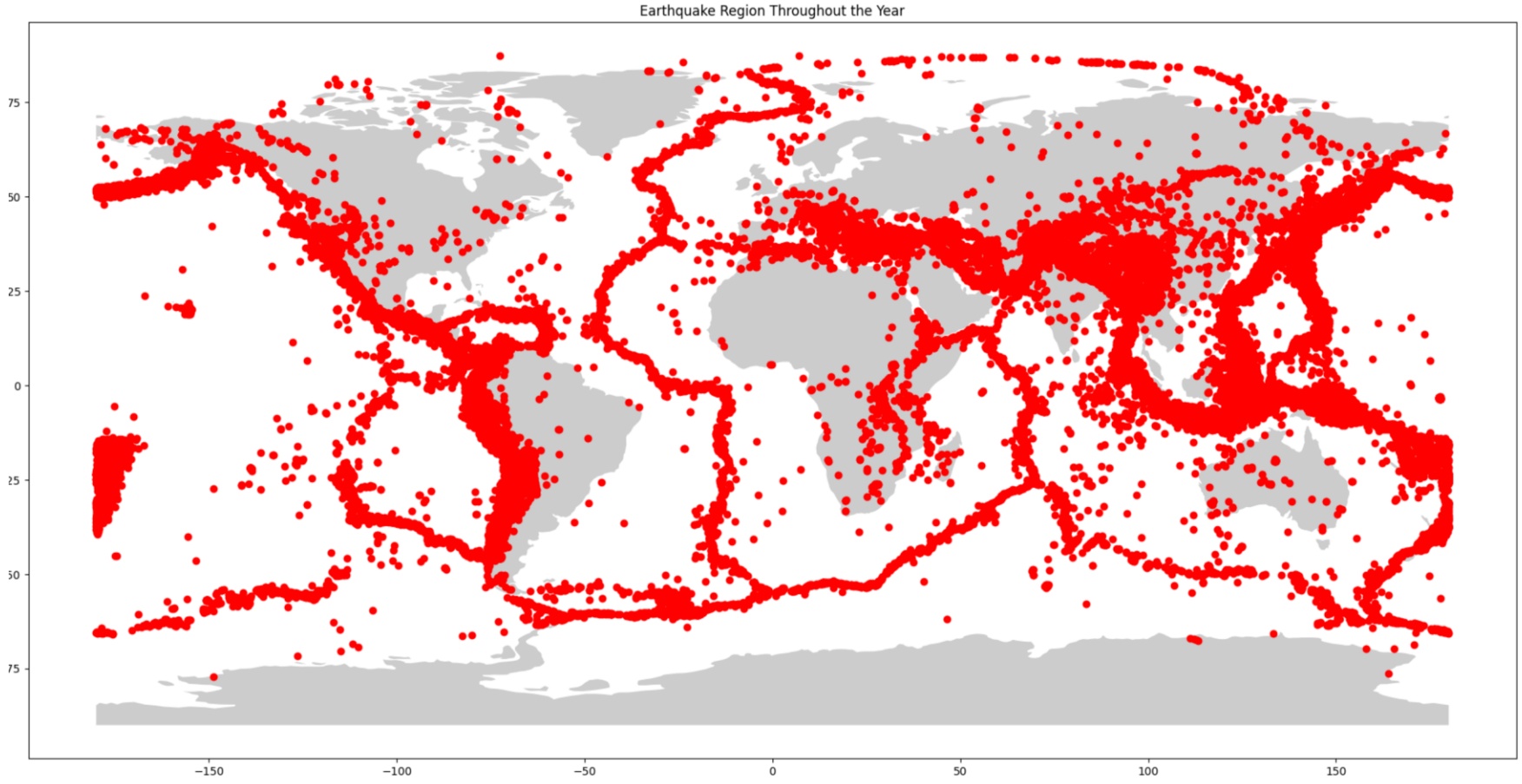
Dataset

```
root
 |-- time: string (nullable = true)
 |-- latitude: double (nullable = true)
 |-- longitude: double (nullable = true)
 |-- depth: double (nullable = true)
 |-- mag: double (nullable = true)
 |-- magType: string (nullable = true)
 |-- nst: double (nullable = true)
 |-- gap: double (nullable = true)
 |-- dmin: double (nullable = true)
 |-- rms: double (nullable = true)
 |-- net: string (nullable = true)
 |-- id: long (nullable = false)
 |-- updated: string (nullable = true)
 |-- place: string (nullable = true)
 |-- type: string (nullable = true)
 |-- horizontalError: double (nullable = true)
 |-- depthError: double (nullable = true)
 |-- magError: double (nullable = true)
 |-- magNst: double (nullable = true)
 |-- status: string (nullable = true)
 |-- locationSource: string (nullable = true)
 |-- magSource: string (nullable = true)
 |-- date: date (nullable = true)
```

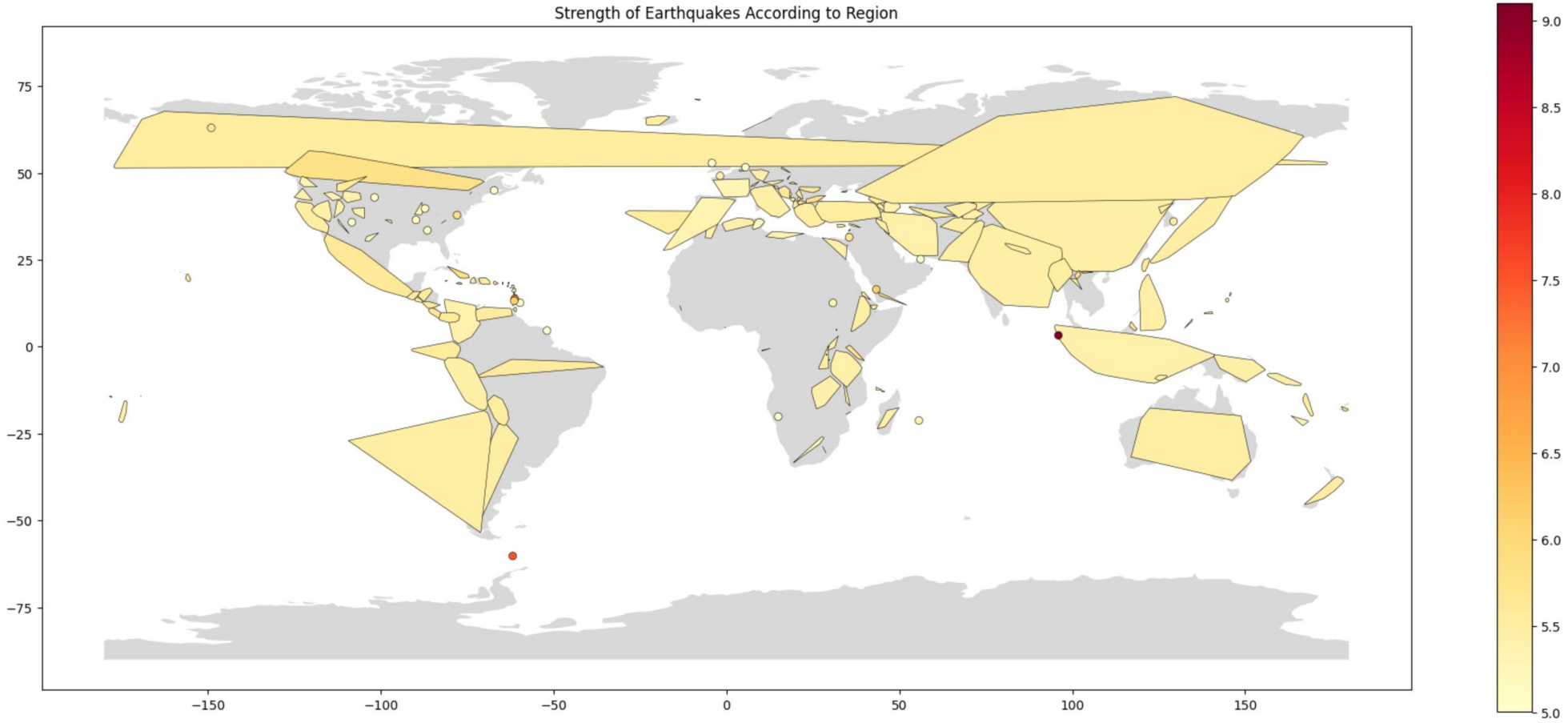
Spatial Analysis

- To discover seismic regions around the earth
- To measure strength of earthquakes according to the region

Spatial Analysis



Spatial Analysis



Graph Analysis – Time Directed Graphs

Vertices – earthquakes

Earthquake I -> Earthquake J are edges if

1. If time(i) < time(j)
2. If time(j) - time(i) <= DELTA_TIME
3. If haversine(i,j) <= DELTA_DISTANCE

$$d = 2r \arcsin\left(\sqrt{\sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)}\right)$$

Graph Analysis - PageRank

date	id	latitude	longitude	mag	depth	place	pagerank
2025-08-20	8589989392	52.539	160.7866	5.0	10.0	156 km ESE of Petropavlovsk-Kamchatsky, Russia	32.34993466358796
2025-08-29	8589989739	52.1139	160.0397	5.1	56.345	142 km SE of Petropavlovsk-Kamchatsky, Russia	21.685158639441212
2025-08-13	8589989427	51.5078	160.0344	5.2	35.0	193 km SE of Vilyuchinsk, Russia	21.60632111324286
2025-08-13	8589988963	51.5078	160.0344	5.2	35.0	193 km SE of Vilyuchinsk, Russia	21.60632111324286
2025-08-13	8589989430	52.1309	160.5079	5.0	10.0	164 km SE of Petropavlovsk-Kamchatsky, Russia	20.805180363453914
2025-08-13	8589988966	52.1309	160.5079	5.0	10.0	164 km SE of Petropavlovsk-Kamchatsky, Russia	20.805180363453914
2025-08-10	8589989449	52.1506	160.417	5.0	10.0	158 km SE of Petropavlovsk-Kamchatsky, Russia	19.698605840451087
2025-08-10	8589988985	52.1506	160.417	5.0	10.0	158 km SE of Petropavlovsk-Kamchatsky, Russia	19.698605840451087
2025-08-11	8589988979	51.7929	160.4078	5.2	10.0	186 km SE of Vilyuchinsk, Russia	15.099542550068923
2025-08-11	8589989443	51.7929	160.4078	5.2	10.0	186 km SE of Vilyuchinsk, Russia	15.099542550068923

Graph Analysis – Triangle Counting

count	date	id	latitude	longitude	mag	depth	place
9611	2025-08-03	8589989812	51.7915	159.8475	5.1	22.905	160 km SE of Vilyuchinsk, Russia
9442	2025-08-03	8589989810	51.5651	159.5352	5.1	10.0	170 km SSE of Vilyuchinsk, Russia
8898	2025-08-03	8589989499	51.4816	159.6145	5.1	10.0	181 km SSE of Vilyuchinsk, Russia
8898	2025-08-03	8589988612	51.4816	159.6145	5.1	10.0	181 km SSE of Vilyuchinsk, Russia
8898	2025-08-03	8589989036	51.4816	159.6145	5.1	10.0	181 km SSE of Vilyuchinsk, Russia
8697	2025-08-02	8589989829	51.638	159.5993	5.2	10.0	165 km SSE of Vilyuchinsk, Russia
8480	2025-08-02	8589989831	51.5932	159.6045	5.4	28.403	169 km SSE of Vilyuchinsk, Russia
8431	2025-08-02	8589988634	51.64	159.5562	5.4	42.574	163 km SSE of Vilyuchinsk, Russia
8431	2025-08-02	8589989058	51.64	159.5562	5.4	42.574	163 km SSE of Vilyuchinsk, Russia
8431	2025-08-02	8589989521	51.64	159.5562	5.4	42.574	163 km SSE of Vilyuchinsk, Russia

THANK YOU!!!